

Introduction to Statistics

Statistics I • Section 1

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Outline

1.1 What Is Statistics?

1.2 Population and Sample

1.3 Data Types and Measurement Levels



What Is Statistics?

Decisions are made without complete information

- A fund manager picks a portfolio → next year's returns are still unknown.
- A firm launches a product → not every buyer was asked.
- We see a **part**, and must speak about the **whole**.

The chain

data → information → decision-making

- Data alone: nothing.
- Data organized so a pattern shows: information.
- Information → decision-making, once we know how far to trust it.

Two modes of the subject

Definition 1.1 (Descriptive and inferential statistics)

1. *Descriptive* (敘述統計): methods that summarize a data set.
2. *Inferential* (推論統計): methods that use a sample to draw conclusions about the population it came from.
 - Average return of *the 30 stocks you hold* → _____, and exactly right.
 - Average of 30 random stocks, used for a market of 900 → _____, and it **can** be wrong.
 - Inference always carries uncertainty, and this course measures it.

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Population and Sample

Population and sample

Definition 1.2 (Population and sample)

1. *Population* (母體): the _____ set of items of interest, size N (possibly infinite).
 2. *Sample* (樣本): an _____ of it, size n .
- The **question** decides the population, not the data you already have.
 - Taiwanese households → one brokerage's clients are **not** the population.

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Why not just measure everyone?

- To know a population number for certain → measure every member: a **census** (普查).
- Usually not possible. Three common reasons:
 - **cost and time**: Taiwan's census runs once every ten years;
 - **measuring destroys the item**: test every light bulb to failure → no bulbs left to sell;
 - **the population cannot be listed at all**.
- So we **sample**: measure a part, speak about the whole.

Can we measure every return?

- What we want: the average daily return of one stock.
- Every return the stock will ever have: that is the census.
- 250 trading days? Ten years? Neither is the full list.
- Still trading → more returns keep coming.

A population with no list

- The returns we saw are only the ones that **did** happen.
- We want what the stock **could** return.
- Such a population: _____ .
- Careful: ten years is not a **bad** sample, just not the whole population.

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Parameter vs. statistic: two very different numbers

Definition 1.3 (Parameter and statistic)

1. *Parameter* (母數/參數): a numerical measure of a **population** characteristic. It is _____, and usually unknown.
 2. *Statistic* (統計量): a numerical measure computed from a **sample**. We know its value once we have the sample, but it _____.
- μ (all 900 stocks): one number, unknown.
 - $\bar{x}$ (your 30 stocks): we can compute it, but it moves.
 - So what we **want** is hidden, and what we **have** is random.

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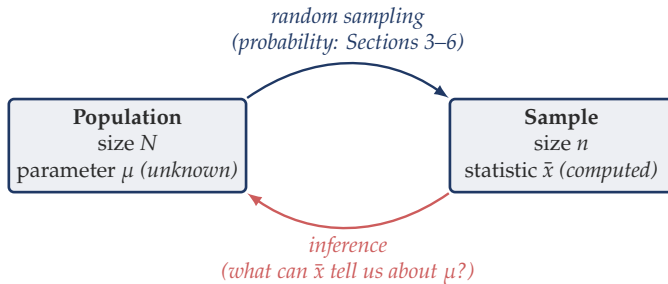
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The inference loop



- Forward arrow: **we** choose the procedure $\rightarrow$ we can model it.
- Backward arrow: only as good as that model.
- Honest question: not “is $\bar{x}$ right?” $\rightarrow$ “how far off can it be?”

How the sample must be drawn

Definition 1.4 (Simple random sample)

A *simple random sample* (簡單隨機樣本) of size n :

1. every member of the population is _____ to be chosen;
 2. choosing one member does not change the chance of choosing another;
 3. every possible sample of size n has the same chance of being chosen.
- It controls the _____, not the result.

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What randomness does and does not promise

- Draw 30 stocks at random from the 900. They could all be large firms.
- Then $\bar{x}$ sits far from μ : the draw was **unlucky**, not wrong.
- Every sample of 30 was equally likely $\rightarrow$ we can compute how often that happens.
- Convenience sample: the stocks on the Yahoo Finance front page.
- Why those? Nobody knows $\rightarrow$ chances unknown $\rightarrow$ nothing to compute.

Systematic sampling (系統抽樣)

- Order the list. Random start among the first j .
- Then take every j th item, where

$$j = \frac{N}{n}.$$

- Cheap, and it usually behaves like a random sample.
- Works only **if** the ordering has nothing to do with the variable we study.
- Danger: a _____ pattern of period j (or dividing j) $\rightarrow$ every item at the same place in that pattern.

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Example 1.5

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A brokerage's client list holds $N = 8,000$ accounts, ordered by the date each was opened. An analyst wants a systematic sample of $n = 200$.

- (a) Find the sampling interval j .
- (b) With random start 7, give the first three accounts selected and the last one.
- (c) Describe one ordering that would make the sample misleading.

Two kinds of error

- **Sampling error** (抽樣誤差): from seeing n items, not all N .
- Unavoidable, _____, and it shrinks as n grows.
- **Nonsampling error** (非抽樣誤差): would remain even in a full _____.
 - **wrong frame**: the list you draw from misses part of the target;
 - **inaccurate response**: unclear, leading or dishonest answers;
 - _____ : you sample those *willing to respond*.

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 - **Nonresponse** (無回應): you sample those *willing to respond*.

Why the difference matters

- Sampling error shrinks toward 0 as n grows.
- A nonsampling **bias** does not.
- So a biased survey of a million is no better than one of a thousand.

Selection bias (選擇偏誤)

- An item enters the sample for a reason connected to _____.
- Sampled population $\neq$ target population.
- A brokerage wants the average return of **all** retail investors, but uses its *premium-research subscribers*.
- Subscribers trade more and hold more $\rightarrow$ their returns are different.
- The frame it sampled from is not the target $\rightarrow$ a huge sample does not remove the bias.

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The everyday version

- Online reviews come from customers who chose to write one.
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- We call this self-selection (自我選擇).

Bias in the time dimension

- Selection bias can also come from **time**.
- A fund database holds the funds that *still exist*.
- Funds that did badly were closed or merged → their returns went with them.
- So the database keeps only the funds that did well, and nothing in the data shows it.
- We call it _____ bias.

Bias in the time dimension

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- A fund database holds the funds that *still exist*.
- Funds that did badly were closed or merged → their returns went with them.
- So the database keeps only the funds that did well, and nothing in the data shows it.
- We call it survivorship (存活者偏誤) bias.
- A bigger sample of survivors is still a sample of survivors.

Example 1.6

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Ten years ago a database tracked 100 equity funds; today 70 still operate and 30 were closed or merged. Over the decade those 70 averaged 8.0% per year; the 30 that disappeared averaged 2.0% per year while they existed. A magazine reports: “the average equity fund returned 8.0% a year.”

- (a) Weight each of the 100 original funds equally, each contributing its own average annual return. What is the mean?
- (b) By how much does the magazine overstate it?
- (c) Which population does 8.0% describe, and why is it useless to an investor today?

Data Types and Measurement Levels

Kinds of variable

Definition 1.7 (Variable; categorical and numerical)

1. *Categorical* (類別變數): values are groups or labels. This holds even when we store them as numeric codes.
 2. *Numerical* (數值變數): quantities from counting or measuring. It is *discrete* (離散) if its values can be listed one by one, *continuous* (連續) if it can take any value in an interval.
- Number of trades placed in a month → _____.
 - Market value of a portfolio → _____.

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The trap: labels stored as numbers

- Satisfaction stored as $1, \dots, 5 \rightarrow$ the entries **are** numerals.
- But “5” is only a label for “very satisfied”. We picked it ourselves, and it could be “A”.
- A rating stored as a number is still a _____ variable.
- The other trap: a share price rounded to two decimals is still continuous.

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Measurement levels

Definition 1.8 (Measurement levels)

1. *Nominal* (名目尺度): labels, no order.
 2. *Ordinal* (順序尺度): we can ____ the values, but no fixed differences.
 3. *Interval* (等距尺度): differences meaningful, zero _____.
 4. *Ratio* (比率尺度): differences meaningful, zero _____.
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 - Interval, ratio $\rightarrow$ *quantitative*.

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Which level is it?

- Nominal, ordinal → the levels of categorical variables.
- Interval, ratio → the levels of _____ ones.
- Test for the last two: is the zero a _____, or does zero mean there is none of the thing?
- 20°C is not twice 10°C . NT\$2m is twice NT\$1m.

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What each level allows

Level	Supports	Does not support	Example
Nominal	counting, mode	ordering, differences	industry sector
Ordinal	ordering, median	differences, means	bond rating
Interval	differences, mean	ratios	temperature (°C)
Ratio	differences, ratios	—	portfolio value

- Read downward: each row supports everything the row above does, and one thing more.
- Averaging a 1–5 scale: the codes carry order, not _____.
- So the mean has no clear meaning → use the median and the frequency distribution.

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- Averaging a 1–5 scale: the codes carry order, not distance.
- So the mean has no clear meaning → use the median and the frequency distribution.

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A bank stores three fields for each of its 500 corporate clients: the credit rating (AAA, AA, ..., C); the calendar year the account was opened; the balance in NT\$. For each computation, say whether it is meaningful, and why.

- (a) The mean credit rating.
- (b) The difference between two clients' opening years.
- (c) The ratio of two clients' balances.
- (d) The ratio of two clients' opening years.